Thapar Institute of Engineering and Technology, Patiala
Electrical and Instrumentation Engineering (EIED)

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

UBM605

Data Structures and Algorithms

AUXILIARY EXAMINATION

BE-BME VI Semester	
Feb 25, 2024	Sunday, 10 AM
Duration: 3 Hrs	Max. Marks: **100**
Name of Faculty :	Dr. Ashish Gupta

General Instructions: All questions are compulsory. Write concise and precise answers. Draw neat, clean, and labeled diagrams wherever required.

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1. (a) Write a recursive algorithm for Mergesort and use the following sequence of keys to demonstrate its working:
   75, 80, 85, 90, 95, 70, 65, 60, 55 **(6+4 marks)**

   (b) Write a procedure/algorithm to convert an infix expression into a postfix expression using stack and demonstrate its working on S+T+U*V*W↑X. **(5+5 marks)**

2. (a) Consider a hash table with 9 slots. The primary hash functions are $h(k) = k \bmod 9$. We insert the following keys in the order 15, 28, 17, 5, 10, 33, 12, 19, 20. Draw the contents of the hash table when the collisions are resolved by
   i) Open hashing.
   ii) Open addressing with Quadratic Probing.
   iii) Closed hashing with Double hashing. Use secondary hash function $h_2(k)$ =7-($k \bmod 7$) **(12 marks)**

   (b) How is a linked list different from an array? Discuss various types of linked-lists with their time-requirements for operations such as insertion at the end, searching and deleting a specific element. **(2+6 marks)**

3. (a) What is a binary search tree (BST)? Suppose the numbers 46 , 45 , 41 , 48 , 43 , 44 , 40 , 49, 56, 42 are inserted in that order into an initially empty BST. The BST uses the usual ordering on natural numbers. Draw the BST starting with an empty tree. Redraw the BST to show the effect of deletion of node with value 41. Finally, traverse the resultant BST using pre-order, in-order, and post-order traversal methods. **(1+2.5+2+3×1.5 marks)**

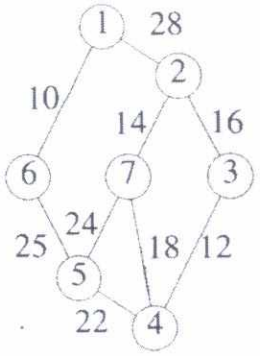


Figure:1

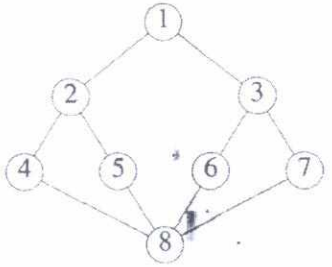


Figure:2

(b) How Prim's algorithm for finding a minimum spanning tree is different from Kruskal's algorithm? Explain the steps of Kruskal's algorithm for computing an MST for the graph shown in Figure 1. **(4+6 marks)**

4. (a) Write an algorithm/pseudocodes for depth first search traversal (DFST) of a graph. Apply the DFST algorithm on node **1** of the graph shown in Figure 2. **(5+5 marks)**

(b) Consider the instance of the string matching problem given in Figure 3.

(i) Compute the LPS array for the Pattern.

(ii) Illustrate the working of KMP algorithm by applying it on the given instance of string matching problem.

(3+7 marks)

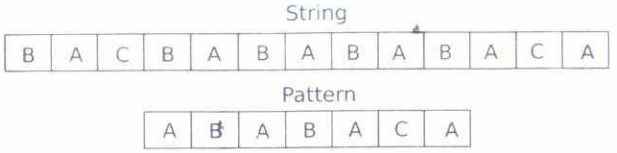


Figure 3

5. Write short notes on the following:

(i) Use of Heaps as priority queue.

(ii) Quick sort

(iii) Circular queues

(iv) Asymptotic notations

(4*5 marks)